12.12 Networking

DynamoDB does not run in your VPC. It is accessible via an API. You need internet connectivity to reach the DynamoDB API. This means you can't access DynamoDB from a private subnet by default, because a private subnet has no route to the internet via an internet gateway. Instead, a NAT gateway is used (see section 5.5 for more details). Keep in mind that an application using DynamoDB can create a lot of traffic, and your NAT gateway is limited to 10 Gbps of bandwidth. A better approach is to set up a VPC endpoint for DynamoDB and use that to access DynamoDB from private subnets without needing a NAT gateway at all. You can read more about VPC endpoints in the AWS documentation at http://mng.bz/51qz.

12.13 Comparing DynamoDB to RDS

In chapter 10, you learned about the Relational Database Service (RDS). For a better understanding, let's compare the two different database systems. Table 12.2 compares DynamoDB and the RDS. Keep in mind that this is like comparing apples and oranges: the only thing DynamoDB and RDS have in common is that both are called databases. RDS provides relational databases that are very flexible when it comes to ingesting or querying data. However, scaling RDS is a challenge. Also, RDS is priced per database instance hour. In contrast, DynamoDB scales horizontally with little or no effort. Also, DynamoDB offers a pay-per-request pricing model, which is interesting for low-volume workloads with traffic spikes. Use RDS if your application requires complex SQL queries or you don't want to